Harsh Kumar

Machine learning engineer (graduating Nov 2026). Deep learning, rigorous evaluation, simulation. I build models against real physics and measure exactly where they break.

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India, open to relocation

BACHELOR THESIS

Accelerating Spatial Universe Simulations: A Performance Analysis of Neural Network Surrogates vs. Traditional Algorithmic Models in Python

Submitted 25 Sep 2026

IU International University of Applied Sciences, Germany

- Built per-body, **variable-N surrogate pipelines in PyTorch** (MLP, LSTM, graph neural networks) for 3D gravitational N-body simulation, trained at body counts N = 10, 25, 50 and 100.
- Validated all three architectures **out of distribution** on real Solar-System initial conditions from **NASA JPL Horizons** ephemerides across 7 presets.
- Designed the evaluation methodology: held-out test splits, unified metrics across model types, and a rollout-stability benchmark quantifying autoregressive error compounding (~5x higher error with feedback loops).
- Published all results, animations and dashboards as a public benchmark site: harsh74-2.github.io/N-body-surrogates.

Projects

agentdrift

github.com/Harsh74-2/agentdrift

- Eval harness for LLM agents: measures how single bad tool calls compound along a trajectory until the final answer breaks. Onset-based per-step error estimation, recovery rates, amplification against independent-error baseline. Pure stdlib Python.
- **Measured** gpt-oss:120b at 56% task success with 13.4% per-step error and 0% recovery: the model never corrects its first mistake.

N-body-surrogates

github.com/Harsh74-2/N-body-surrogates

- Open benchmark: MLP, LSTM and GNN compared under identical data pipelines, training budgets and unified evaluation, with per-N result pages and animation galleries.

VR Web App

github.com/Harsh74-2/VR-WEB-APP

- Interactive virtual-reality experience in the browser. JavaScript and WebGL, no native install.

Skills

ML / DL	PyTorch, graph neural networks, LSTM / RNN, MLP, training pipelines, OOD validation
Python	NumPy, data pipelines (npz / json), matplotlib, testing
Engineering	Git, Linux (WSL2), reproducible experiments, code review
Web / Other	JavaScript, Three.js / WebGL, HTML / CSS, LaTeX

Education

B.Sc. Computer Science

April 2022 – November 2026

IU International University of Applied Sciences, Germany

- Coursework focus: machine learning, algorithms, software engineering, mathematics.